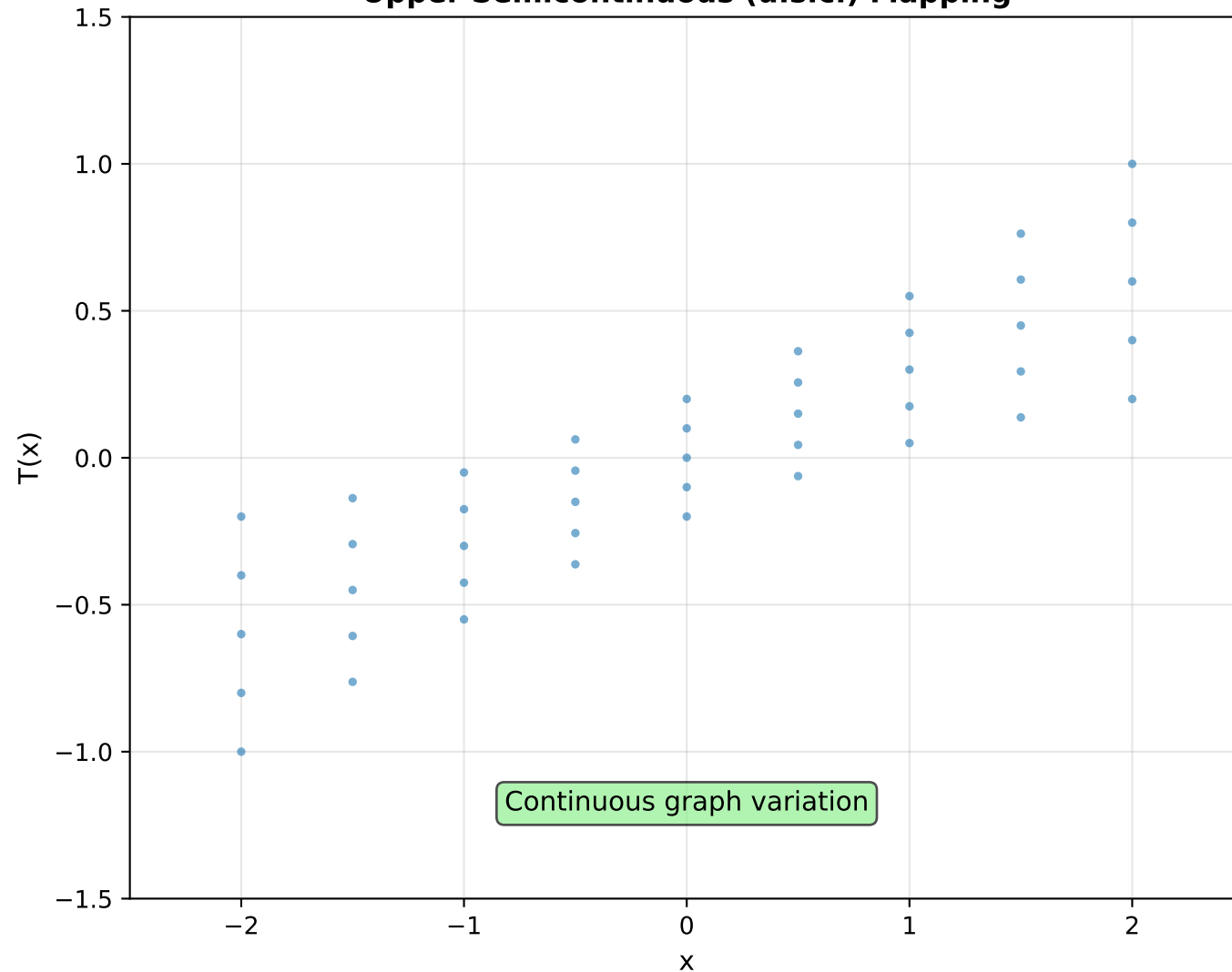


Upper Semicontinuous (u.s.c.) Mapping



Definition of Upper Semicontinuity

$T: X \rightarrow 2^Y$ is u.s.c. if for each closed $B \subseteq Y$,
the set $T^{-1}(B) = \{x: T(x) \cap B \neq \emptyset\}$ is closed

Equivalently: For each $x \in X$ and open set $U \supseteq T(x)$,
there exists open $V \ni x$ such that $U \supseteq T(y)$ for all $y \in V$